

PA00FG

The Gauss–Eisenstein Data Lemma

A standard mathematical restatement of
`distinct_odd_primes_gauss_eisenstein_data_exists`

Formal status. The native tactic body is kernel-checked relative to its declared dependencies, but the theorem is currently a `candidate_body_checked` result rather than a publicly admitted closed theorem.

1 Abbreviations for the expanded PA formulas

All variables range over $\mathbb{N}$. The native theorem uses only 0, successor, addition, multiplication, equality, first-order connectives, and quantifiers. The familiar notation below abbreviates formulas that are fully expanded on the theorem page.

The PA primality predicate used in the statement is

$$\text{Prime}_{\text{PA}}(r) :\iff r \neq 1 \wedge \forall a, b \in \mathbb{N}, (r = ab \implies a = 1 \vee b = 1). \quad (1)$$

Quadratic residuosity is encoded without subtraction:

$$\text{QRes}_m(a) :\iff \exists x, u, v \in \mathbb{N}, \quad x^2 + mu = a + mv. \quad (2)$$

Thus $\text{QRes}_m(a)$ is the natural-number formulation of $x^2 \equiv a \pmod{m}$.

Parity is expressed by

$$\text{Even}(n) :\iff \exists t \in \mathbb{N}, \quad n = 2t, \quad (3)$$

$$\text{Odd}(n) :\iff \exists t \in \mathbb{N}, \quad n = 2t + 1. \quad (4)$$

Finally, balanced congruence modulo two means

$$a \equiv b \pmod{2} :\iff \exists u, v \in \mathbb{N}, \quad a + 2u = b + 2v. \quad (5)$$

2 The clean mathematical statement

Theorem 1 (PA00FG). *Let $p, q, h, k \in \mathbb{N}$. Suppose*

$$p = 2h + 1, \quad q = 2k + 1, \quad \text{Prime}_{\text{PA}}(p), \quad \text{Prime}_{\text{PA}}(q), \quad p \neq q. \quad (6)$$

Then there exist $e, f, Q, U \in \mathbb{N}$ such that

$$\text{QRes}_p(q) \iff \text{Even}(e), \quad (7)$$

$$\neg \text{QRes}_p(q) \iff \text{Odd}(e), \quad (8)$$

$$\text{QRes}_q(p) \iff \text{Even}(f), \quad (9)$$

$$\neg \text{QRes}_q(p) \iff \text{Odd}(f), \quad (10)$$

$$e \equiv Q \pmod{2}, \quad (11)$$

$$f \equiv U \pmod{2}, \quad (12)$$

$$Q + U = hk. \quad (13)$$

Because $p = 2h + 1$ and $q = 2k + 1$, the parameters are

$$h = \frac{p-1}{2}, \quad k = \frac{q-1}{2}. \quad (14)$$

The equations in (6) are used instead of subtraction or division because neither operation is primitive in the native PA term language.

3 What the four witnesses represent

The proof constructs e and f as the two Gauss-lemma sign counts. In ordinary notation, one may view them as

$$e = \# \left\{ 1 \leq i \leq h : \langle iq \rangle_p > \frac{p}{2} \right\}, \quad (15)$$

$$f = \# \left\{ 1 \leq j \leq k : \langle jp \rangle_q > \frac{q}{2} \right\}, \quad (16)$$

where $\langle n \rangle_m$ denotes the least positive residue of n modulo m . Gauss’s lemma explains (7)–(10): quadratic residuosity is exactly the evenness of the appropriate sign count.

The proof chooses Q and U as the Eisenstein quotient sums

$$Q = \sum_{i=1}^h \left\lfloor \frac{iq}{p} \right\rfloor, \quad (17)$$

$$U = \sum_{j=1}^k \left\lfloor \frac{jp}{q} \right\rfloor. \quad (18)$$

The parity bridges (11) and (12) identify the Gauss counts with these quotient sums modulo two. Eisenstein’s lattice-point argument gives the exact identity

$$\sum_{i=1}^h \left\lfloor \frac{iq}{p} \right\rfloor + \sum_{j=1}^k \left\lfloor \frac{jp}{q} \right\rfloor = hk. \quad (19)$$

Formal nuance. The public surface of PA00FG exposes only the relations (7)–(13). The much larger internal proof constructs the beta-coded finite sequences and sums that realize (15)–(18), then hides those implementation witnesses before returning e, f, Q, U .

4 Why this is the central quadratic-reciprocity package

In conventional Legendre-symbol notation, the four classification formulas say

$$\left(\frac{q}{p} \right) = (-1)^e, \quad \left(\frac{p}{q} \right) = (-1)^f. \quad (20)$$

The two parity bridges and the Eisenstein identity therefore give

$$\begin{aligned} \left(\frac{q}{p} \right) \left(\frac{p}{q} \right) &= (-1)^{e+f} \\ &= (-1)^{Q+U} \\ &= (-1)^{hk} \\ &= (-1)^{\frac{(p-1)(q-1)}{4}}. \end{aligned} \quad (21)$$

Equation (21) is the standard parity identity behind quadratic reciprocity. PA00FG is therefore not yet the final reciprocity theorem; it is the common witness package from which the final same-status and opposite-status cases are derived constructively.